

Toward an AI Maturity Model in Healthcare: Identifying Core Dimensions and Critical Success Factors

Karlo Filipovic, Rosch Said, Luka Saban, Mustafa Uzdilli, Mike Krey
Zurich University of Applied Science, School of Management and Law, Winterthur, Switzerland
filipkar@students.zhaw.ch, muhamros@students.zhaw.ch, sabanluk@students.zhaw.ch,
uzdilmus@students.zhaw.ch, Mike.Krey@zhaw.ch

Abstract

Artificial Intelligence is increasingly recognized as a critical enabler for transforming hospital operations and improving healthcare delivery. However, the absence of healthcare-specific maturity models limits the systematic adoption of AI in clinical settings. This study addresses this gap by conducting a structured literature review of existing AI maturity models and critical success factors across domains. The analysis identifies six core dimensions: technology, data, strategy, people, organization, and regulations. These findings highlight the multifaceted nature of AI integration and underscore the need for a tailored approach in complex healthcare environments. By providing a conceptual foundation, this work advances the development of future AI maturity models to support healthcare leaders in assessing AI readiness, ensuring strategic alignment, and facilitating structured AI integration within hospital settings. Further empirical validation is needed to refine the framework for practical application.

Keywords: AI maturity model, maturity assessment, artificial intelligence, AI readiness, hospital, healthcare

1. Introduction

Artificial intelligence (AI) has emerged as a transformative technology across various sectors, enabling data-driven decision-making, process automation, and adaptive system behavior [30]. Based on this cross-sectoral potential, AI is gaining increasing relevance in healthcare, particularly in hospital settings, where it is seen as a strategic response to growing systemic pressures [1]. In view of rising healthcare costs, workforce shortages, demographic shifts, and the growing demand for personalized care, AI is now considered a key enabler in the advancement of modern hospital systems [7].

AI offers a broad range of potential applications within hospitals, including support in diagnostics and treatment planning, process optimization, and enhanced

administrative efficiency [32]. In medical care, AI supports radiologists by analyzing X-rays, CT scans, and MRIs more accurately and quickly [6]. It helps detect diseases at earlier stages, reducing misinterpretations and speeding up diagnosis. AI tools can highlight anomalies that might be missed by the human eye, thereby improving diagnostic precision. This results in faster clinical decisions and better treatment outcomes. Ultimately, AI enhances patient safety and contributes to more effective healthcare delivery.

AI-powered virtual assistants provide real-time health support for patients, such as medication reminders and appointment scheduling [32]. They can monitor chronic conditions and offer personalized health advice, improving daily self-care and adherence to treatment plans. These systems help reduce unnecessary doctor visits by promoting proactive health management. By offering accessible support, they also improve the overall quality of life for patients and enhance continuity of care. In administration, AI improves hospital efficiency by optimizing scheduling, admissions, and resource allocation [27]. It analyzes large datasets to forecast demand, prevent bottlenecks, and streamline operations. This allows staff to focus more on patient care by automating administrative tasks. Reduced waiting times, better use of resources, and lower costs are some of the key outcomes. As a result, both operational performance and patient satisfaction increase. It turned out, that the structured and responsible implementation of AI is not merely a technical concern but also a societal imperative, particularly in addressing the widening gap between clinical demand and available healthcare resources [13].

1.1 Problem Statement & Objectives

Despite its potential, the large-scale adoption of AI in hospitals remains limited due to a range of interrelated challenges [19]. These include high implementation costs, limited relevant data, and misalignment with institutional strategies [38]. Many barriers are human-

centric, such as lack of expertise, resistance to change, and concerns related to trust, ethics, and accountability [38]. Technical, operational, and regulatory constraints further hinder integration, often intensified by the absence of standardized implementation frameworks [26].

This structural gap leads to institutional fragmentation, reduces interoperability, and complicates the incorporation of AI into clinical workflows [19]. In order to support the structured implementation of AI in complex environments such as hospitals, maturity models have become an increasingly valuable tool. Maturity models are defined as an analytical instrument that characterizes the behaviors, practices, and processes enabling an organization to achieve consistent and sustainable outcomes [49]. It constitutes a reference framework that facilitates a more precise understanding of organizational realities, with the dual purpose of explaining phenomena and supporting predictive assessments. Comparable to architectural models, which serve to illustrate the complexities of construction rather than to execute it, maturity models provide conceptual guidance rather than direct implementation. These models help organizations assess their current state of AI integration, identify development stages, and guide strategic planning. They offer a structured way to reflect on technological, organizational, and regulatory readiness for AI adoption.

A key component of such models is the use of Critical Success Factors (CSFs). These factors represent concrete elements that are essential for the successful implementation of AI, such as data quality, leadership support, or ethical guidelines. Within the maturity model, CSFs serve as practical criteria to evaluate progress across different areas of an organization. The maturity model provides an overarching framework, while the CSFs offer a more detailed view of what is required at each stage of development. This study combines both approaches and aims to analyze existing AI-related maturity models, identify relevant dimensions and CSFs, and synthesize these insights into a conceptual foundation for the development of an AI maturity model tailored to hospital settings. To address the current lack of structured guidance, this contribution proposes an initial theoretical framework that serves as a basis for future research and empirical validation in hospital contexts.

The previously described research motivation and subsequent research problem allow for the formulation of the following research question (RQ):

RQ: “How can the maturity of Artificial Intelligence (AI) in hospitals be systematically supported?”

In order to gain deeper perceptions of related issues, answering the research question encompasses the following accompanying knowledge-oriented goals:

- “To systemize existing AI maturity models”

- “To analyze which dimensions and CSFs are relevant for assessing AI maturity in hospitals?”

The paper is structured into several main sections. First, the methodology is outlined, including the approach to the literature review and the reclassification of heterogeneous terms into six overarching dimensions. This section also explains the role of CSFs and their methodological integration. The following part introduces the hospital environment and regulatory frameworks, providing the conceptual foundation for understanding AI adoption in healthcare.

Subsequently, a comparative analysis of existing AI maturity models is presented, highlighting recurring dimensions and patterns across different frameworks. Building on this, the identified CSFs are synthesized and systematically assigned to the dimensions. The discussion section then addresses methodological, regulatory, and ethical implications of AI adoption in hospitals and situates the findings in an international context. Finally, the conclusion summarizes the main contributions, outlines limitations, and suggests avenues for future research.

2. Methodology

This study adopts a structured literature review to address the RQs. Given the absence of AI maturity models explicitly developed for the healthcare sector, the review synthesizes existing models from various domains to identify transferable dimensions relevant to hospitals.

The search strategy was informed by systematic review principles and guided by Cooper’s taxonomy [10], as illustrated in Table 1.

The taxonomy distinguishes six key characteristics: (1) focus, (2) goal, (3) perspective, (4) coverage, (5) organization, and (6) audience. The literature review was conducted across Scopus, Web of Science, IEEE Xplore, ACM Digital Library, and PubMed, covering the publication period from 2019 to 2025. These databases were selected to ensure comprehensive coverage of multidisciplinary, technical, and healthcare-relevant studies. The search terms were applied to both titles and abstracts. The review followed a two-step approach. The first step focused on AI maturity models. A targeted search using the terms (“AI” OR “Artificial Intelligence”) AND (“maturity model” OR “maturity assessment” OR “maturity level”) identified 586 publications: Scopus (297), Web of Science (109), IEEE Xplore (150), PubMed (21), and ACM Digital Library (9).

A multi-stage screening process was applied, starting with title and abstract review to exclude irrelevant studies, followed by full-text screening based on predefined criteria: peer-reviewed publications, frameworks for AI maturity or capability assessment, and organizational

focus. This process resulted in 21 studies included for analysis.

Table 1. Literature review framework based on Cooper (p. 109) [10]

Category	AI Maturity Models	Critical Success Factors
(1) Focus	Theories and practices or applications: Review of AI maturity models as theoretical frameworks and their use in various domains.	Practices or Applications: Review of empirical studies and case analyses identifying CSFs for AI implementation in healthcare, public sector and organizations.
(2) Goal	Integration & Generalization: Synthesis of common structures, dimensions, and patterns across models.	Integration & Generalization: Synthesis of common CSFs across different studies and domains.
(3) Perspective	Neutral: Objective comparison without endorsing any specific model.	Neutral: Objective synthesis of findings from the selected studies.
(4) Coverage	Exhaustive with Selective Citation: Systematic screening of 586 papers, with 21 studies included based on predefined eligibility criteria.	Representative: Systematic screening of 111 papers, with 11 studies included based on predefined eligibility criteria.
(5) Organization	Historical: Chronological arrangement of models by publication year.	Conceptual: CSFs were thematically synthesized by dimension across multiple studies, with sources cited directly alongside each CSF.
(6) Audience	Specialized Scholars & Practitioners: Specialized scholars in AI and digital transformation, as well as healthcare decision-makers such as hospital managers, IT leads, and digital strategy officers.	

The second step addressed CSFs for AI implementation in healthcare, organizations, and the public sector. A targeted search using the terms (“AI” OR “Artificial Intelligence”) AND (“Critical Success Factors” OR “CSFs”) AND (“Healthcare” OR “Hospitals” OR “Organizations”) identified 111 publications: Scopus (72), Web of Science (15), IEEE Xplore (19), PubMed (4),

and ACM Digital Library (1). Screening followed the same process and criteria as the first step, resulting in 11 studies included for final analysis.

To address the heterogeneity of terms and concepts identified in the reviewed literature, a structured reclassification process was conducted. The goal was to consolidate diverse and often inconsistently used constructs (e.g., *governance*, *leadership*, *ethics*, *compliance*) into a coherent set of six overarching dimensions: Technology, Data, Strategy, People, Organization, and Regulations. The reclassification followed a coding scheme based on thematic analysis [68]. All extracted terms were first coded according to their semantic content and contextual usage in the respective sources. In a second step, codes were grouped by conceptual similarity and aligned with one of the six dimensions. For instance, terms such as *data availability*, *data quality*, and *data governance* were subsumed under the Data dimension, while *leadership commitment* and *managerial support* were assigned to People. Ambiguous cases were resolved through iterative discussion among the authors to ensure intersubjective validation.

The inclusion criteria for terms were defined as follows [69]: (1) only peer-reviewed publications were considered, (2) explicit reference to AI maturity models, critical success factors (CSFs), or comparable frameworks was required, and (3) the term had to be situated in an organizational or healthcare-related context. This ensured that the coding process remained both rigorous and relevant to the scope of the study.

3. AI in the Hospital Environment

The research problem is approached from two principal perspectives: 1) the hospital environment, characterized by its specific conditions, characteristics, and constraints, and 2) the domain of AI, encompassing a broad spectrum of technologies, regulations and conceptual framework. This chapter therefore delineates the conceptual foundation that informs the subsequent analysis of AI maturity models in healthcare.

3.1 Hospital Environment in Switzerland

To enhance clarity and provide a concrete point of reference, the Swiss healthcare system is used as an illustrative example in this study. Its federal and decentralized structure offers valuable insights into the interaction between political frameworks, administrative responsibilities, and hospital organization. While the specifics reflect the Swiss context, many of the described dynamics and challenges can be generalized and applied to healthcare systems in other countries.

Healthcare in Switzerland is shaped by its political and administrative frameworks, which have evolved alongside the historical development of the Swiss Confederation [51]. The federal system is organized into three levels of government: (1) the confederation, (2) the cantons, and (3) the municipalities. Each level assumes distinct responsibilities for health policy, resulting in a system that is both federal and decentralized [52].

Growing cost pressures and the increasing complexity of processes and technologies are among the key drivers of continuous reorganization within Swiss hospitals. Yet, the integration of processes is often impeded by the traditional tripartite division of hospital functions into medical care, nursing, and administration [54]. From an external perspective, a hospital appears as a unified organization with the overarching goal of delivering high-quality care. Internally, however, the relationships among stakeholders, the underlying power structures, and the question of who effectively governs the hospital are less transparent [55].

Hospitals remain strongly influenced by a functional separation of medical disciplines. On the one hand, this departmentalization allows for clear task allocation and efficiency in urgent situations. On the other hand, it restricts flexibility, creating uncertainty when key decision-makers are unavailable. On average, about 80% of hospital employees in Switzerland work in medical or nursing roles [6], which gives these groups substantial influence over organizational culture.

This is also reflected in professional relationships. The study by Lau and colleagues has noted the persistence of a “silo mentality” in healthcare [56]. Grahmann and Gutwetter [57] further describe tensions between physicians and nurses as a historically rooted rivalry, shaped by hierarchical distinctions and unequal distributions of prestige, influence, and decision-making power [59]. The dominance of the medical profession is reinforced by managerial responsibilities and authority, often positioning nursing as a supportive but undervalued counterpart [58].

3.2 AI and Regulatory Frameworks

The domain of AI is extensive and integrates a broad spectrum of disciplines, including logic, probability theory, continuous mathematics, perception, reasoning, learning, and decision-making [6].

For the purpose of this study, AI is defined as the capability of computer systems to execute tasks that traditionally require human intelligence. This encompasses the ability to learn from data and prior experiences, to recognize and interpret patterns, to make autonomous decisions, and to continuously improve performance over time through data-driven analysis and adaptive mechanisms [60].

The integration of AI in healthcare, particularly in the form of Machine Learning (ML) applications, is heavily shaped by regulatory frameworks.

In Europe, two major legal instruments determine the conditions for the approval and use of AI-based medical software: 1) the Medical Device Regulation (MDR) [61] and 2) the EU AI Act [62]. Together, these frameworks address both the general requirements for medical devices and the specific challenges posed by AI technologies.

The MDR, in force since 2017, classifies software-based medical devices into four risk categories. Depending on the potential clinical impact, products range from self-certification for low-risk applications to comprehensive conformity assessments for high-risk systems, such as AI-based diagnostic tools [63]. While the MDR ensures patient safety and product quality, it also introduces substantial complexity for AI applications, especially regarding explainability and lifecycle management, which remain underdeveloped in the MDR framework.

The EU AI Act, legally binding since 2025, complements these requirements by focusing specifically on AI-related risks. It introduces a four-tier risk classification for AI systems, ranging from minimal to unacceptable risk, with corresponding obligations for transparency, documentation, and human oversight [62]. High-risk systems, including many diagnostic and treatment-supporting AI tools, must undergo rigorous risk assessments and conformity evaluations. A key challenge arises from the absence of an exemption for in-house developed systems, which means that hospitals creating and deploying their own AI solutions are considered “providers” and must meet the same requirements as commercial manufacturers. The AI Act thereby extends the regulatory scope and raises the compliance burden, especially for smaller providers.

In contrast, the United States regulates medical devices through the Food and Drug Administration (FDA), which adopts a centralized, risk-based, and relatively flexible approach [64]. Products are divided into three

risk classes and may follow one of four approval pathways: Premarket Notification (510(k)), Premarket Approval (PMA), Investigational Device Exemption (IDE), or Humanitarian Device Exemption (HDE). Especially relevant for AI is the FDA's concept of Software as a Medical Device (SaMD) and its lifecycle approach, introduced in 2021 [65]. This framework allows continuous learning systems to be updated under predefined conditions without requiring full re-approval, through mechanisms such as Predefined Change Control Plans (PCCP).

The FDA model thus combines rigorous safety standards with adaptive mechanisms for innovation, offering a contrast to the more rigid EU framework.

A direct comparison of MDR and FDA regulations highlights fundamental differences. The MDR relies on decentralized "Notified Bodies" and imposes stringent documentation and clinical validation requirements, which may hinder rapid innovation, particularly for SMEs [61]. The FDA, by contrast, concentrates authority within one agency, providing consistent oversight but allowing more flexibility, especially for AI-driven systems [64].

Overall, both systems aim to safeguard patients and ensure product effectiveness, yet they pursue these objectives through different institutional structures and regulatory philosophies. For AI in healthcare, this results in considerable compliance challenges. In Europe, the combined demands of MDR and the AI Act increase regulatory complexity, particularly regarding transparency, explainability, and data governance. In the U.S., the FDA framework, though demanding in terms of evidence and quality assurance, offers a more innovation-friendly environment for adaptive and learning AI systems. For globally operating companies, a deep understanding of both regimes is essential to ensure lawful market access and to balance patient safety with the pace of technological innovation.

In the healthcare domain, the implementation of AI requires not only compliance with regulatory frameworks but also the integration of Explainable AI (XAI) in order to safeguard transparency, accountability, and reliability in critical decision-making processes [66]. XAI emphasizes the need for transparency across all stages of AI development, including system design, data management, and validation procedures, thereby enabling stakeholders to comprehend and evaluate the reasoning behind AI-generated outcomes.

In clinical practice, the application of third-party AI solutions to complex medical cases makes it essential that the underlying decision-making logic of the system is transparent. This enables healthcare professionals to interpret and critically assess AI recommendations, fostering trust and facilitating their effective integration into patient care [67].

3.3 Spread of AI in the Hospital Environment

A survey, conducted in spring 2023 among 79 general hospitals in Switzerland (response rate: 57%, N=45), examined the knowledge, current use, benefits, and barriers related to AI [50]. Responses were supplemented by ten expert interviews from the fields of medical device manufacturing, consulting, and AI research.

Overall, the findings reveal that the baseline knowledge of AI among hospital IT managers remains limited. Some respondents incorrectly associated unrelated concepts such as "e-learning" or "blockchain" with AI, indicating conceptual uncertainty.

With respect to current applications, speech recognition systems were most frequently mentioned, followed by radiology-related solutions. High-cost technologies such as surgical robots were reported only in large hospitals, while care robots were not in use. Strikingly, a considerable number of respondents indicated "no AI solutions," which, according to experts, may reflect unawareness of existing applications rather than their actual absence. Additional use cases included IT security systems with embedded AI features.

The perceived benefits of AI were largely concentrated on improvements in diagnostic quality, efficiency gains in patient treatment, and enhanced support for second opinions. Expectations included improved organizational efficiency in patient care, but not necessarily increased physician-patient time [50].

The barriers to adoption were predominantly financial, both in terms of acquisition and operation. Interestingly, "none of the listed reasons" was cited with equal frequency, which experts attributed to factors such as lack of expertise and technical shortcomings. Regulatory requirements, project management complexity, and resource constraints were also noted as obstacles.

Importantly, the survey found no significant differences in AI knowledge or use between small and large hospitals. Awareness and application levels were comparable across hospital sizes. Generative AI was not yet a salient topic among IT managers at the time of the survey. Concerns regarding job losses or distrust in AI scored surprisingly low, likely due to the IT orientation of the respondents; surveys involving medical personnel might yield different perspectives [50].

Finally, the issue of regulatory compliance emerged as ambivalent. While regulations such as the Medical Device Regulation (MDR) were broadly recognized as necessary, they were simultaneously perceived as hindering innovation and the introduction of new AI solutions. Moreover, responsibility for ensuring compliance of AI applications remained unclear. Notably, many hospitals seemed unaware that even self-developed AI applications fall under regulatory scrutiny and must undergo conformity assessment.

The Swiss findings resonate strongly with the international landscape. Global analyses, such as Yao and Yu's survey of LLM-based multi-agent AI hospitals (2023–2025) [53], demonstrate that hospitals worldwide face comparable dynamics: AI deployment remains concentrated in speech recognition, radiology, and well-defined diagnostic support tools, whereas more complex applications such as robotics or agent-based hospital ecosystems are still largely experimental. The expected benefits such as diagnostic accuracy, efficiency gains, and enhanced decision support are consistently emphasized across countries, but financial constraints, lack of technical expertise, and regulatory uncertainty remain universal barriers [53]. Importantly, the emergence of LLM-driven multi-agent hospital simulations illustrates that international research is moving toward integrated, system-level AI solutions capable of automating entire clinical workflows, supporting education, and generating synthetic training data. These trends highlight that the challenges identified in Switzerland are not isolated but represent broader structural issues in aligning innovative AI technologies with healthcare systems worldwide. Although hospitals recognize the necessity for change and first improvements can be observed, the concept of AI has not yet been consistently established. Empirical findings suggest that private hospitals, in particular, show greater awareness of the strategic integration of AI, underlining the importance of IT as a central management challenge [50]. The limited diffusion of AI principles in the hospital context can be attributed to four overarching barriers, which at the same time represent the most critical obstacles to successful convergence of business and IT:

1. **Lack of expertise.** Hospitals often face deficits in methodological and technical knowledge, as well as limited access to structured support for the implementation of AI principles.
2. **Lack of resources.** Budgetary restrictions, staff shortages, and the prioritization of clinical processes over administrative or AI-related initiatives constrain the ability to implement AI effectively.
3. **Lack of motivation.** The perceived benefits of AI are frequently judged as insufficient compared to the required investments. This perception, reinforced by structural characteristics of the hospital market such as guaranteed public funding and limited competitive pressure (Switzerland), results in resistance at both the individual and organizational levels.
4. **Lack of reputation of IT.** IT is still often viewed primarily as a cost center. Business priorities and the complexity of clinical processes dominate, which leads to IT receiving limited strategic attention and visibility.

In summary, the analysis highlights that while AI in healthcare holds considerable promise, its adoption remains constrained by regulatory complexity, limited expertise, and organizational barriers. Against this backdrop, the use of maturity models offers a systematic way to assess readiness, structure implementation processes, and guide hospitals in overcoming these challenges.

4. AI maturity assessment in the hospital Environment

Originally developed in the context of software engineering (e.g., the Capability Maturity Model), maturity models have since been adapted to a wide range of domains, including healthcare, digital transformation, and organizational development [42]. Such models serve as diagnostic and strategic tools, enabling institutions to assess their current level, identify gaps and plan further actions in alignment with strategic goals [3].

In healthcare settings, they are particularly valuable for navigating complex and interdisciplinary innovations such as AI, where progress depends not only on technology but also on cultural, regulatory, and structural factors [7].

In the context of AI adoption in hospitals, maturity models can help ensure that implementation efforts are not purely experimental or isolated but embedded in a broader organizational strategy that includes data governance, ethical oversight, workforce readiness, and sustainable infrastructure [26]. Hospitals deal with disjointed circumstances that impede systematic integration, such as strict data protection regulations, issues with the explainability of AI models, and infrastructure-related limitations [7].

By providing organized, step-by-step instructions suited to hospital-specific needs, a maturity model can help overcome these hurdles and form the backbone of structured transformation processes which are crucial for bridging the gap between pilot projects and scalable clinical integration [7]. AI has the potential to enhance care delivery, streamline processes, and assist medical personnel in all hospital tasks [32]. However, to fully realize these advantages, systematic execution, regulatory alignment, and widespread professional acceptability are necessary in addition to technological readiness [30].

4.1 Structuring AI readiness through CSFs

CSFs serve as a foundational term for identifying what must be in place for AI adoption to succeed [5]. Rather than being abstract principles, CSFs represent concrete, context-sensitive requirements derived from

empirical studies, expert judgment, and strategic priorities [45]. They provide focus and direction, especially in hospital settings, where legal, procedural, and ethical complexities make AI implementation particularly challenging [30]. In the context of AI maturity models, CSFs do more than support implementation, they often define the structure of the model itself [39].

These factors help decision-makers identify readiness gaps, allocate resources effectively, and reduce the risk of failure during transformation [43]. Especially in healthcare, where fragmented IT systems and patient safety concerns dominate, CSFs act as navigational tools [45]. They allow organizations to focus on areas that are not just beneficial but essential, where failure would undermine the entire initiative [14]. As such, CSFs are both analytical lenses and operational levers for managing AI adoption [14].

5. AI Maturity Models

To establish a foundation for an AI maturity assessment in hospitals, this section presents a comparative analysis of existing AI maturity models. The focus lies on identifying key dimensions and patterns across frameworks from various domains. Table 2 provides an overview of selected AI maturity models, summarizing their application areas, number of maturity levels, specific naming conventions for maturity stages, and core dimensions. The analysis reveals that the dimensions **Strategy**, **Organization**, **Data**, **Technology**, **People** and **Regulations** are the most frequently cited and thus represent the core components across diverse frameworks. Regulations is also mentioned, but typically in varying forms or embedded within other dimensions. These dimensions consistently emerge as fundamental for assessing AI readiness and capability development in organizational contexts. For developing an AI maturity model tailored to hospitals, the most frequently cited dimensions should serve as the conceptual foundation, as they capture the key factors critical for AI adoption in complex hospital environments.

The dimensions identified in this study correspond closely to those delineated by Duncan et al. [16], who conceptualize digital maturity in hospitals through core dimensions including strategy, technological capability, governance, interoperability, patient-centered care, human resources, and data analytics. While not fully identical, there is thematic alignment: Strategy and Technology correspond directly, Organization covers governance, management, and interoperability, Data aligns with Data Analytics, and People reflects Skills and Behavior. Regulations is identified as a distinct dimension in this work.

6. Critical Success Factors

Building on the dimensions identified through comparative analysis, this section synthesizes the Critical Success Factors (CSFs) that operationalize these dimensions. CSFs represent the essential conditions that must be fulfilled in order to ensure the successful adoption and sustained use of AI in hospitals. They translate abstract dimensions into actionable requirements and provide guidance for hospital management and clinical stakeholders seeking to build AI readiness. Each CSF is assigned to one of the six dimensions, reflecting its thematic and operational focus.

Technology. Hospitals require reliable and scalable technical infrastructures to enable the deployment of AI systems within complex clinical environments. CSFs in this dimension include robust hospital IT architectures that integrate seamlessly with existing hospital information systems (HIS), electronic health records (EHRs), and radiology information systems (RIS). Adequate computing power is necessary for tasks such as the real-time processing of imaging data or predictive modeling of patient outcomes (see Section 3.3). Technical reliability and system resilience are critical to avoid downtime in emergency and intensive care units, where uninterrupted service is vital.

Data. Data quality and accessibility are central for hospital-based AI solutions. CSFs include the structured collection of patient data across departments (e.g., diagnostics, laboratory, radiology), the implementation of hospital-wide data governance frameworks, and strict compliance with data protection standards such as GDPR (see Section 3.1). To ensure fair and accurate treatment recommendations, datasets must be representative of the hospital's patient population and free from systematic bias. Hospitals must also guarantee that data exchange across departments and with external partners (e.g., laboratories, insurers) is secure and efficient.

Strategy. Strategic alignment is necessary to ensure that AI projects contribute to the hospital's mission of improving patient care and operational efficiency. CSFs include a clear hospital-wide AI vision, commitment from the hospital board and medical leadership, and sustainable financing models for AI infrastructure. Strategic foresight also requires hospitals to anticipate regulatory changes (e.g., EU AI Act, MDR) that affect the deployment of AI-driven diagnostics and treatment support (see Section 3.2). Embedding AI into long-term hospital strategies ensures that projects are not isolated pilots but integrated into the overall healthcare delivery system.

Table 2. Overview of AI maturity models

Author	Application Area	Levels	Maturity Levels	Dimensions
Fornasiero et al. (2025)	AI and Big Data maturity in process industry	4	Initial, Experimenting, Formalising, Optimising	Strategy, Organisation, People, Technology, Data
Ferreira et al. (2025)	Responsible AI adoption in organizations across industries	4	Unaware, Exploratory/Reactive, Proactive, Strategic	Human Agency, Technical Robustness, Transparency, Diversity, Societal/Environmental Wellbeing, Accountability, Privacy
Kudryavtsev et al. (2025)	AI Implementation Capability in SMEs	5	No AI usage, Basic AI use, AI models with configuration, AI models developed, AI systems developed, deployed, and maintained	AI Model Development, Data Management, AI Use Cases, Regulatory & Ethical Risks, Infrastructure, Change Management
Eliseo et al. (2025)	AI Adoption in Industry 4.0	6	Not Applicable, Initial, Alert, Acceptable, Ideal, Optimal	Strategy, People, Process, Technology, Customers, Manufacturing & Operations, Products & Services, Information Technology, Culture
Hansen et al. (2024)	AI Capability Diffusion in Organizations	6	Inactive, Explorative, Ad-hoc, Formalized, Embedded, Transformational	Technological (Data, Infrastructure), Organizational (Strategy, People, Culture), External (Ethics/Regulation, Motivation)
Sonntag et al. (2024)	AI Deployment Capability in Manufacturing	5	Initial, Experimental, Practicing, Integrated, Transformed	Culture & Competencies, Strategy, Data, Organization & Processes, Technology
Dreyling et al. (2024)	AI maturity in public sector organizations	5	Initial, Assessing, Determined, Managed, Optimized	Technology, Data, People & Competencies, Organization & Processes, Leadership & Strategy, Financial, Law/Ethics/Trust, Security & Privacy
Das et al. (2024)	Trustworthy AI System Development	4	Basic, Establishing, Managed, Reliable	4 Core Pillars (Auditability, Explainability, Fairness, Safety) × 3 AI Lifecycle Dimensions (Data Ingestion & Featurization, Modeling, Operationalization)
Shakeripour & Ronaghi (2024)	AI Maturity in Animal Farming	4	AI Novice, AI Ready, AI Proficient, AI Advanced	Strategy, Organization, Data, Technology, Operations
Shabou (2024)	AI-Enhanced Appraisal Processes in Archives & Records Management	5	Not Existing, Manual/Ad-hoc, Reproducible, Systematic, Automated	Data Governance, AI Tool Integration, Explainability, Ethical Considerations, Process Integration
Costa & Mendonça (2024)	AI Policy Maturity in Organizations	4	Initial, Developing, Established, Optimized	Data Governance, Compliance, Transparency, Ethical AI, Risk Management, AI Strategy, Management Support
Graibeh et al. (2024)	AI and Blockchain Adoption Capability in Public Sector	5	Initial, Developed, Defined, Managed, Optimized	Strategy & Governance, Technology, People, Process
Cho et al. (2023)	Trustworthy AI Software Development Processes	6	Incomplete, Performed, Managed, Established, Predictable, Optimizing	Plan & Design, Data Collection & Processing, Model Development & Evaluation, System Evaluation, Operation & Monitoring, Fairness & Safety-specific processes
Noymanee et al. (2022)	AI Maturity in Government Administration and Service	5	New Entry, Elementary Entry, Operational, Proficiency, Advanced	Strategy, Organization, Data, Technology
Ramdhani & Surendro (2022)	AI Governance Maturity in Enterprises	6	Incomplete, Initial, Managed, Defined, Quantitatively Managed, Optimizing	Data Layer, Technology Layer, Ethical & Social Layer, Legal Layer
Bhat et al. (2022)	Industrial AI Maturity Assessment	5	Exploration, Experimentation, Stabilization, Expansion, Transformation	Strategy, Governance, People, Process, Technology (Input, Process, Output), Data
Schuster & Waidelich (2022)	AI Capability Maturity in SMEs with focus on Privacy & Ethics	5	Beginner, Basic, Intermediate, Advanced, Expert	Strategy, Organization, Culture/Mindset, Technology, Data, Privacy, Ethics
Colangelo et al. (2022)	AI Adoption in Smart Production Planning and Control (PPC)	5	Initialization, Definition, Preparation, Conditionally Intelligent, Fully Intelligent	Strategy, Data, Organization, Technology
Yams et al. (2020)	AI Innovation Capability in Organizations	5	Foundational, Experimenting, Operational, Inquiring, Integrated	Strategy, Ecosystem, Mindsets, Organization, Data, Technology, Trustworthiness (Ethics, Legality, Robustness)

Lichtenthaler (2020)	AI Management Capability in Organizations	6	Isolated Ignorance, Initial Intent, Independent Initiative, Interactive Implementation, Interdependent Innovation, Integrated Intelligence	Strategy, Integration, Coordination, Human-AI Collaboration, Innovation
Ellefsen et al. (2019)	AI Readiness in Logistics 4.0	5	AI Novice, AI Ready, AI Proficient, AI Advanced, AI Expert	Strategy, Organization, Data, Technology, Operations

Table 3. CSFs for AI Adoption in Hospitals by Dimension

Dimension	ID	Critical Success Factors	References
Technology	T1	Technical infrastructure is scalable and flexible to support future AI growth and complexity.	[5], [14], [21], [33], [34]
	T2	Technology platforms and tools are orchestrated in alignment with business needs and workflows.	[34], [46]
	T3	Systems are adaptable and capable of handling real-time or large-volume data processing for AI tasks.	[34], [46]
	T4	There is internal or external access to deep AI knowledge and expertise.	[14], [43]
	T5	Legacy systems must be modernized or replaced to enable smooth integration with AI technologies.	[35], [46]
	T6	Existing IT infrastructure must be evaluated and upgraded to support AI integration into network operations.	[35], [46]
	T7	Standardization of digital processes and data formats is needed to ensure AI system integration.	[14]
Data	D1	The organization ensures high-quality, consistent, and relevant data is available for AI training and operations.	[5], [33], [34], [39], [46]
	D2	Data from internal and external sources is integrated into a unified infrastructure accessible for AI purposes.	[5], [14], [34], [46]
	D3	All data is labelled and regularly validated to support supervised machine learning.	[14], [34], [46]
	D4	A formal data governance framework is in place, covering ownership, access rights and use of data.	[14], [34], [46]
	D5	Data is stored, distributed, and made accessible across departments as needed for AI use cases	[14], [34], [39], [46]
	D6	Data processing capabilities (e.g., integration, cleaning, real-time handling) are in place to support AI deployment.	[14], [34], [43], [46]
Strategy	S1	A regulatory framework is co-developed with stakeholders to govern liability, consent, and transparency.	[5], [26], [39], [46]
	S2	AI initiatives must be aligned with clear long-term organizational goals	[5], [33], [35], [39], [46]
	S3	Financial planning is in place to manage the total cost of AI implementation and scaling.	[14], [33], [35], [39]
	S4	Top management actively supports AI initiatives and promotes AI as a strategic priority.	[14], [35], [39], [43], [46]
	S5	AI efforts are aligned with existing values, strategic goals, and operational practices.	[14], [39], [46]
People	P1	Aligning AI deployment with measurable operational performance outcomes.	[15], [47]
	P2	Defined project management and oversight for organizational AI initiatives.	[42]
	P3	Necessary resources (budget, staff, tools) are allocated for AI success.	[34], [47]
	P4	A culture of innovation and learning supports AI experimentation.	[35]
	P5	A clear AI governance structure is defined with roles, policies, and ethical oversight mechanisms.	[15]
	P6	Continuous feedback mechanisms are established to adapt AI practices and foster learning.	[15], [44]
	P7	Practical competence to use AI tools in diagnostics and analysis.	[8]

Organization	O1	Aligning AI deployment with measurable operative performance outcomes.	[2]
	O2	Defined project management and oversight for organization AI initiatives.	[42]
	O3	Necessary resources (budget, staff, tools) are allocated for AI success.	[34], [47]
	O4	A culture of innovation and learning supports AI experimentation.	[35]
	O5	A clear AI governance structure is defined with roles, policies, and ethical oversight mechanisms.	[15]
	O6	Continuous feedback mechanisms are established to adapt AI practices and foster learning.	[15], [44]
Regulations	R1	AI systems are auditable, with lifecycle documentation and traceability embedded in model development.	[12]
	R2	Explainability is ensured through mechanisms like model cards, feature tracking, and interpretability modules.	[12]
	R3	Fairness and bias mitigation are integrated into AI development, from data collection to prediction outputs.	[12]
	R4	Cultural and legal context is addressed through adaptive regulation and clear liability mechanisms.	[2]
	R5	Internal AI policy frameworks support regulation, including ethical governance and algorithm registries.	[15], [25]

People. Hospital staff are central to AI readiness. CSFs in this dimension include continuous training of clinicians, nurses, and administrative staff to understand and effectively use AI systems. Acceptance among healthcare professionals is fostered through participatory implementation, where medical staff are actively involved in the design, testing, and evaluation of AI solutions (see Section 3.1). Interdisciplinary collaboration between doctors, nurses, IT specialists, and administrators is essential to embed AI into clinical workflows. Hospitals must also establish clear communication and change management strategies to address potential resistance and to promote trust in AI-assisted decision-making.

Organization. Hospitals are highly complex organizations with hierarchical structures and siloed departments. CSFs in this dimension include well-defined governance mechanisms to coordinate AI initiatives across medical and administrative units, as well as process maturity to adapt workflows for AI integration (see Section 3.1). Interdepartmental collaboration is especially critical for applications such as AI-assisted surgery scheduling, which requires input from anesthesiology, surgery, and bed management. Hospitals must also develop organizational routines for monitoring AI performance and integrating feedback into clinical and administrative processes.

Regulation. Hospitals operate in one of the most strictly regulated sectors. CSFs in this dimension include compliance with the MDR and EU AI Act in Europe or FDA regulations in the U.S., particularly for AI-based diagnostic tools and decision-support systems (see Section 3.2). Hospitals must ensure that every AI application, whether procured from external vendors or developed in-house, undergoes appropriate conformity

assessments. In addition to legal compliance, hospitals must establish internal guidelines addressing ethical concerns such as transparency, accountability, and fairness in patient treatment. Regulatory adherence thus safeguards patient trust and institutional credibility while minimizing legal and clinical risks.

Taken together, these hospital-specific CSFs illustrate that successful AI adoption is not solely dependent on technological innovation but requires alignment with clinical workflows, organizational structures, staff acceptance, and strict regulatory compliance. Table 3 summarizes the dimensions, their categorized CSFs, and corresponding references.

To assess the prevalence of the identified six dimensions across the reviewed maturity models, a frequency analysis was conducted. Each occurrence of a concept associated with one of the six dimensions (Technology, Data, Strategy, People, Organization, Regulations) was coded and counted. Table 4 provides an overview of the results.

The results highlight that **Technology** and **Data** are most prominently represented, reflecting their role as foundational enablers for AI implementation. Without adequate technological infrastructure and high-quality data, AI adoption in hospitals cannot be realized effectively. **Strategy** and **People** dimensions are also well represented, pointing to the importance of leadership commitment, staff engagement, and alignment of AI with organizational goals. By contrast, **Regulations** are considerably less developed within the existing models. In most cases, regulatory aspects are only mentioned implicitly, for example in relation to data protection or compliance, rather than treated as a standalone category. This indicates a significant gap in current maturity models, especially given the increasing impact of MDR and

the EU AI Act on AI deployment in healthcare (see Section 3.2)

Table 4: Frequency of Dimensions in Reviewed AI Maturity Models (n=21)

Dimension	Frequency (n)	Relative Prevalence (%)
Technology	18	86%
Data	17	81%
Strategy	14	67%
People	13	62%
Organization	11	52%
Regulations	6	29%

The justification for category formation is twofold. First, the six dimensions emerged as recurrent conceptual clusters across the analyzed models, reflecting both technological and organizational determinants of AI adoption in healthcare. Second, their aggregation provides analytical clarity by reducing redundancy and capturing conceptual overlaps.

To further substantiate this classification, a frequency analysis was conducted, indicating how often each dimension appeared across the 21 reviewed models. Preliminary findings reveal that dimensions such as Technology and Data are more prominently represented, reflecting the technological focus of most AI maturity approaches, whereas Regulations is comparatively underdeveloped. This observation underscores the necessity of explicitly integrating regulatory aspects into maturity models for healthcare AI, where compliance and patient safety are central.

7. Discussion

This paper aimed to investigate which factors are relevant for assessing AI maturity in hospitals, focusing on the most important dimensions and CSFs as outlined in research question and accompanied research goals. The structured literature review revealed, while several AI maturity models exist across various domains, there is a notable absence of a comprehensive, domain-specific framework tailored specifically to the unique needs of hospitals.

The review identified six key dimensions: technology, data, strategy, people, organization, and regulations. These dimensions appeared consistently across the reviewed maturity frameworks. Together, they represent both technical and non-technical prerequisites for AI integration. Moreover, they reflect the complex interplay between infrastructure readiness, strategic alignment, regulatory constraints, and human-centered factors such as workforce competencies and institutional acceptance. These dimensions were compared with the

hospital-focused digital maturity dimensions of Duncan et al. [16] to ensure conceptual alignment and contextual relevance.

The CSFs identified in this review point to persistent implementation challenges. These include fragmented data ecosystems, insufficient data governance mechanisms, organizational resistance to change, and limited institutional leadership in advancing digital transformation [34]. These findings support the argument that AI adoption in hospitals cannot be approached solely through a technical perspective. Instead, successful integration requires a holistic and multi-level coordination that considers organizational, legal, and ethical factors in parallel with technological performance.

Beyond methodological and regulatory considerations, the adoption of AI in healthcare raises important ethical and sociopolitical questions that must be addressed to ensure responsible implementation.

Bias. AI systems are highly dependent on the quality and representativeness of their training data. If the underlying datasets are biased, whether through demographic underrepresentation, historical inequalities, or incomplete records, the resulting models risk reproducing or amplifying existing disparities. In healthcare, such biases could lead to systematic disadvantages for specific patient groups, potentially undermining the principle of fairness in medical care.

Trust. The trustworthiness of AI systems is a prerequisite for their acceptance by both medical professionals and patients. Black-box models that fail to provide transparent or interpretable decision-making processes may hinder clinical adoption. Approaches such as Explainable AI (XAI) can play a crucial role in building trust, as they enable physicians to critically assess and contextualize AI-generated recommendations before integrating them into treatment decisions.

Equity. The use of AI also poses questions of equity and accessibility. If advanced AI systems are only available in well-funded hospitals or high-income regions, a widening gap in the quality of care could emerge. Moreover, inequities can arise if AI-based tools perform differently across patient populations, for example when predictive accuracy varies by ethnicity, gender, or socioeconomic background. Addressing these risks requires not only robust validation but also explicit policy measures to promote equitable access.

Patient Harm. Ultimately, insufficiently validated or poorly monitored AI systems have the potential to cause direct harm to patients. Errors in diagnosis or treatment recommendations, particularly in high-stakes clinical settings, can have severe consequences. Regulatory safeguards such as the MDR, FDA frameworks, and the EU AI Act provide essential mechanisms to mitigate such risks, yet their effective enforcement depends

on rigorous testing, ongoing monitoring, and clear assignment of accountability. Taken together, these considerations underscore that the governance of AI in healthcare cannot be limited to technical and regulatory dimensions alone. A comprehensive perspective that incorporates ethical and sociopolitical implications is necessary to ensure that AI contributes to improved patient outcomes while safeguarding fundamental values such as fairness, trust, and patient safety.

Building on the identified methodological, regulatory, and ethical considerations, future research should focus on the empirical validation and refinement of the proposed dimensions and CSFs. Several avenues appear particularly promising:

1. **Empirical Case Studies.** In-depth investigations in diverse hospital settings are required to test the applicability and practical utility of the six dimensions. These studies should assess how well the model captures the complexities of AI adoption in real-world environments.
2. **Expert Evaluation.** Structured interviews and Delphi studies with stakeholders from healthcare, technology, and regulatory bodies can provide critical feedback on the model's completeness, accuracy, and usability.
3. **Integration of Ethical Dimensions.** Further research should explicitly incorporate ethical and sociopolitical factors such as bias, equity, and patient safety into the model, thereby extending its scope beyond technical and organizational readiness.

By pursuing these directions, future studies can ensure that the maturity model is not only conceptually robust but also empirically grounded, adaptable to diverse healthcare contexts, and aligned with ethical and regulatory imperatives.

8. Conclusion

The primary aim of this research was to establish a structured foundation for assessing AI maturity in hospitals. This goal was addressed by systematically reviewing existing maturity models and extracting transferable CSFs from a wide range of domains. The resulting foundation integrates six key dimensions: technology, data, strategy, people, organization, and regulations, which were consistently identified as relevant. These dimensions represent the technical, strategic, and institutional prerequisites that must be fulfilled to support the successful and responsible adoption of AI in hospitals. The conceptual foundation contributes to the current state of research in two ways. First, it offers a consolidated framework for evaluating AI readiness in a sector where such structured guidance is currently

lacking. Second, it operationalizes the concept of maturity by associating each dimension with specific CSFs enabling a more granular and actionable assessment process.

From a conceptual perspective, the results can serve as a reference point for the further refinement of AI maturity constructs specific to hospitals. Future research should focus on the empirical validation of the model through case studies, expert interviews, or quantitative assessments in real-world clinical settings. Such validation would not only strengthen the model's credibility but also enable the development of benchmarking instruments or self-assessment tools that could be used across hospitals.

Furthermore, longitudinal studies could help capture how AI maturity evolves over time and under which conditions certain dimensions become critical. As healthcare systems worldwide continue to digitize and face pressures related to cost-efficiency, personalization, and workforce shortages, the ability to systematically assess and improve AI maturity will become increasingly important [7].

9. Limitations

Although the review process was conducted in a systematic and rigorous manner, several limitations must be considered. One of the main limitations is the weak availability of AI maturity models that were specifically developed for the healthcare sector. Therefore, the conceptual foundation presented in this study is primarily informed by maturity frameworks originating from other domains, including manufacturing, public services, and enterprise IT. While these models contribute valuable structural insights, their applicability to hospitals may be constrained by contextual differences, particularly regarding clinical workflows, regulatory requirements, and data governance challenges. Another limitation concerns the heterogeneity of the sources from which the CSFs were derived. These sources varied in methodological rigor and contextual specificity, which introduces the possibility of either underrepresenting healthcare-specific challenges or overgeneralizing findings from non-healthcare contexts. Furthermore, the conceptual foundation identified in this study has not yet been validated through empirical application in clinical settings. As a result, its practical effectiveness and operational relevance remain to be demonstrated.

10. References

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